

An elementary obstruction for

$$y^3 + xy = x^4 + x + 3$$

The equation has **no integer solutions**. The obstruction comes from two norm factorizations: they force two positive odd integers to be congruent to 1 or 3 modulo 8, although their product is congruent to 7 modulo 8.

1 An elementary prime-divisor lemma

Lemma 1. *If p is a prime with $p \equiv 5$ or $7 \pmod{8}$ and $p \mid a^2 + 2b^2$, then $p \mid a$ and $p \mid b$. Consequently, if a positive odd integer has no prime divisor congruent to 5 or 7 modulo 8, it is congruent to 1 or 3 modulo 8.*

Proof. We first show that $r^2 = -2$ in $\mathbb{F}_p$ implies $p \equiv 1$ or $3 \pmod{8}$. Choose a root u of $T^2 - rT - 1$ in a field containing $\mathbb{F}_p$. Such a field is either $\mathbb{F}_p$ itself or the quotient by this quadratic if it is irreducible. Then

$$u^2 = ru + 1, \quad u^3 = r - u, \quad u^4 = -1.$$

Thus u has multiplicative order exactly 8. The two roots of the quadratic are u and u^3 . In characteristic p , the binomial theorem and $r^p = r$ give

$$(u^p)^2 - ru^p - 1 = (u^2 - ru - 1)^p = 0.$$

Here $r^p = r$ is elementary: multiplication by any nonzero residue permutes the nonzero residues, and multiplying them proves $r^{p-1} = 1$. Hence $u^p = u$ or $u^p = u^3$, so $p \equiv 1$ or $3 \pmod{8}$.

Now, if $p \nmid b$ and $a^2 + 2b^2 \equiv 0 \pmod{p}$, then $r = ab^{-1}$ would satisfy $r^2 = -2$, a contradiction. Thus $p \mid b$, and then $p \mid a$. Finally, a positive odd integer with no forbidden prime divisor is a product of primes congruent to 1 or 3 modulo 8; such a product is again 1 or 3, since $3^2 \equiv 1 \pmod{8}$. $\square$

2 Nonexistence theorem

Theorem 1. *There is no pair $(x, y) \in \mathbb{Z}^2$ satisfying*

$$y^3 + xy = x^4 + x + 3. \tag{1}$$

Proof. Suppose that (1) holds.

Positivity and parity. For every integer x , one has $x^4 + x + 3 \geq 3$. If $y \leq 0$, positivity of the left side would force $x = -u$, $y = -v$ with $u, v > 0$ and $u > v^2$. In particular $v \leq u - 1$, giving

$$y^3 + xy = v(u - v^2) \leq u(u - 1) < u^4 - u + 3 = x^4 + x + 3,$$

where the strict inequality follows from $u^4 - u^2 + 3 > 0$. This is impossible. Also $y = 1$ would give $x^4 = -2$. Thus $y > 1$.

Modulo 2, equation (1) gives $y(1 + x) \equiv 1$, so x is even and y is odd. Modulo 8, therefore,

$$(1 + x)y \equiv x + 3 \pmod{8}.$$

Every odd residue is its own inverse modulo 8, and for even x one has $x^2 \equiv 2x \pmod{8}$. It follows that

$$y \equiv (1+x)(x+3) \equiv 2x+3 \pmod{8}. \quad (2)$$

The first factor. Set

$$D = y^2 + y + 1 + x.$$

Equation (1) gives

$$(y-1)D = x^4 + 2 = (x^2)^2 + 2 \cdot 1^2. \quad (3)$$

Since $y > 1$, we have $D > 0$; it is also odd. If a prime $p \equiv 5$ or $7 \pmod{8}$ divided D , Lemma 1 applied to $x^4 + 2$ would imply $p \mid 1$, a contradiction. Consequently,

$$D \equiv 1 \text{ or } 3 \pmod{8}. \quad (4)$$

The second factor. Define

$$\ell = 2x - 13, \quad C = y^2 + y\ell + \ell^2 + x, \quad A = x^2 - 4x + 20, \quad B = 7x - 30.$$

The key identity is

$$(y-\ell)C = x^4 + x + 3 - \ell^3 - x\ell = A^2 + 2B^2. \quad (5)$$

Moreover, C is odd, and it is positive because

$$12C = 3(2y + 2x - 13)^2 + 4(3x - 19)^2 + 77 > 0. \quad (6)$$

We claim that C has no prime divisor $p \equiv 5$ or $7 \pmod{8}$. Indeed, such a divisor would divide $A^2 + 2B^2$, and Lemma 1 would force $p \mid A$ and $p \mid B$. But

$$49A = B(7x + 2) + 1040, \quad 1040 = 2^4 \cdot 5 \cdot 13, \quad (7)$$

so $p = 5$ or $p = 13$.

For $p = 5$, the condition $p \mid B$ gives $x \equiv 0 \pmod{5}$, hence $\ell \equiv 2 \pmod{5}$. Therefore $p \mid C$ would imply

$$0 \equiv C \equiv y^2 + 2y + 4 = (y+1)^2 + 3 \pmod{5},$$

so $(y+1)^2 \equiv 2 \pmod{5}$, impossible since the square residues are 0, 1, 4.

For $p = 13$, the condition $p \mid B$ gives $x \equiv 8 \pmod{13}$, hence $\ell \equiv 3 \pmod{13}$. Thus $p \mid C$ would imply

$$0 \equiv C \equiv y^2 + 3y + 4 \pmod{13}, \quad (2y+3)^2 \equiv 6 \pmod{13}.$$

This too is impossible: the square residues modulo 13 are 0, 1, 3, 4, 9, 10, 12. The claim is proved, and Lemma 1 now gives

$$C \equiv 1 \text{ or } 3 \pmod{8}. \quad (8)$$

The contradiction. Since $\ell \equiv 2x + 3 \equiv y \pmod{8}$ and y is odd,

$$D \equiv 3x + 5 \pmod{8}, \quad C \equiv x + 3 \pmod{8}.$$

The four possible residues of the even integer x give

$x \pmod{8}$	0	2	4	6
$D \pmod{8}$	5	3	1	7
$C \pmod{8}$	3	5	7	1

No column is consistent with both (4) and (8). Equivalently, $DC \equiv 7 \pmod{8}$, whereas a product of two residues from $\{1, 3\}$ must lie in $\{1, 3\}$. This contradiction proves the theorem. $\square$

3 Motivation for the construction

The useful starting point is not to compare the sizes of y^3 and x^4 , but to exploit the shared term $xy - x = x(y - 1)$. Subtracting the value at $y = 1$ produces (3). The expression $x^4 + 2$ is of the form $a^2 + 2b^2$ with $b = 1$, so it cannot have a prime divisor congruent to 5 or 7 modulo 8. This forces the first odd factor D into the residues 1, 3 and, by the last table, restricts x to 2, 4 modulo 8.

A second factorization must exclude precisely those remaining residues. Equation (2) suggests subtracting a linear value $\ell = 2x + b$ with $b \equiv 3 \pmod{8}$. Its complementary factor $y^2 + y\ell + \ell^2 + x$ is then automatically $x + 3$ modulo 8, exactly the needed opposite restriction.

To construct a norm identity, seek integers u, v, w, b such that

$$x^4 + x + 3 - (2x + b)^3 - x(2x + b) = (x^2 - 4x + u)^2 + 2(vx + w)^2.$$

The coefficient -4 is forced by the cubic term. Matching the remaining coefficients gives

$$\begin{aligned} u + v^2 &= -6b - 9, \\ -8u + 4vw &= 1 - b - 6b^2, \\ u^2 + 2w^2 &= 3 - b^3. \end{aligned}$$

The tuple $(b, u, v, w) = (-13, 20, 7, -30)$ satisfies these three identities and has the required $b \equiv 3 \pmod{8}$. It therefore produces

$$x^4 + x + 3 - (2x - 13)^3 - x(2x - 13) = (x^2 - 4x + 20)^2 + 2(7x - 30)^2.$$

The two norm coordinates can share a forbidden odd prime only at 5 or 13, as (7) shows, and both exceptions are eliminated by the displayed square-residue checks. Thus the second factor also has only the allowed prime divisors. The decisive step is the combination of these two norm restrictions, not a congruence test on the original equation alone.